

SOFTWARE ENGINEERING LABS

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Municipal Infrastructure Damage Reporting App

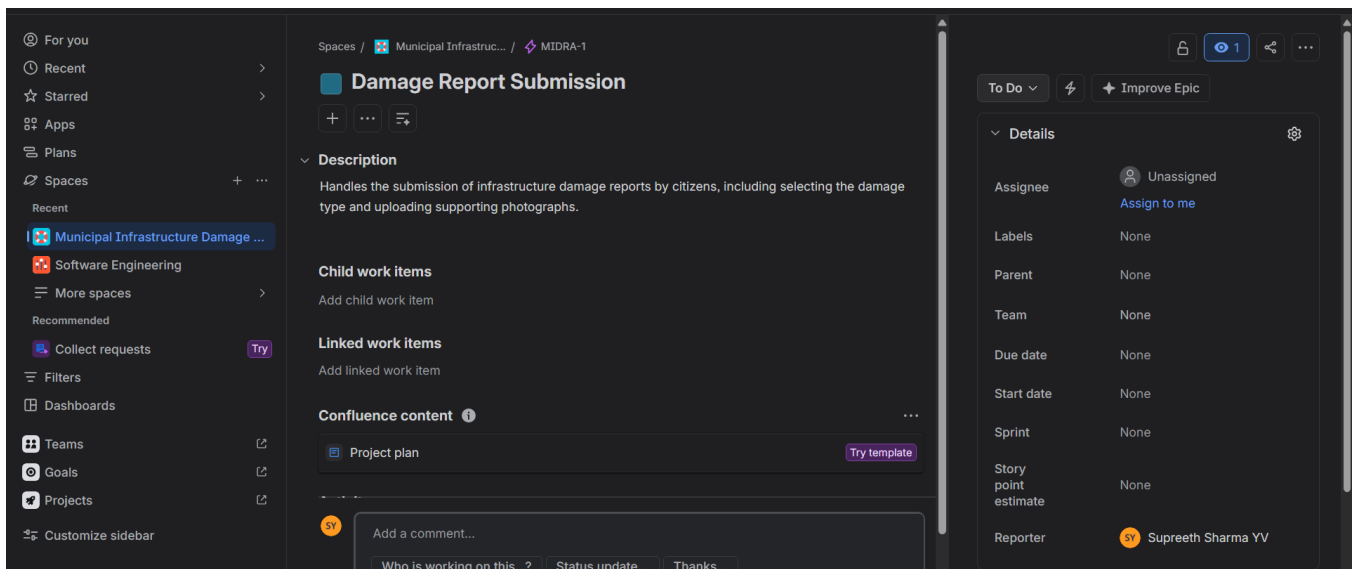
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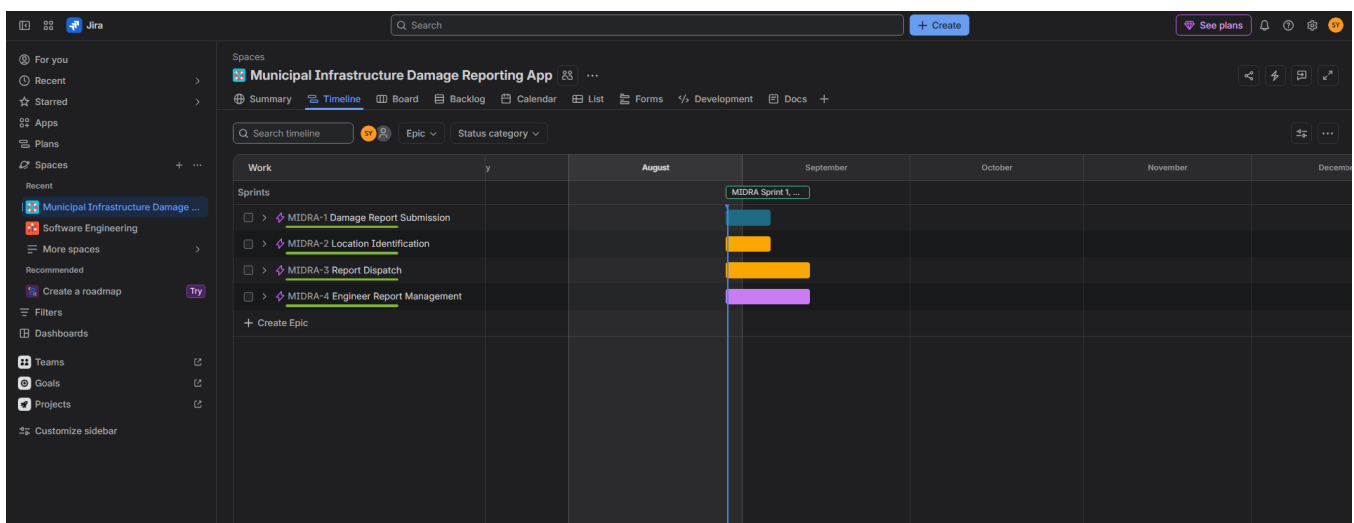
Section: 5th Semester K Section

EPICS

The project was organized into four major epics covering the main municipal damage-reporting workflow: MIDRA-1 Damage Report Submission, MIDRA-2 Location Identification, MIDRA-3 Report Dispatch, and MIDRA-4 Engineer Report Management. Together, these epics represent the citizen reporting flow and the subsequent processing and management of a damage report.



Jira Timeline



STORIES AND STORY POINT ASSIGNMENTS

The backlog contains five user stories distributed across the four epics. Story points represent relative effort. Sprint 1 was planned with 10 points, while Sprint 2 contained the remaining 8 points.

Work item	Story	Epic	Points
MIDRA-5	Submit Damage Report	Damage Report Submission	3
MIDRA-6	Upload Damage Photo	Damage Report Submission	2
MIDRA-7	Extract GPS Location	Location Identification	5
MIDRA-8	Dispatch Damage Report	Report Dispatch	5
MIDRA-9	Manage Damage Report	Engineer Report Management	3

Jira Backlog

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MIDRA Sprint 1

Add dates

0 work items

Start sprint

Plan your sprint

Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.

Create

0 of 0 work items visible

Estimate: 0 of 0

Backlog

5 work items

Create sprint

Create a whiteboard to plan your work

MIDRA-5 Submit Damage Report

MIDRA-6 Upload Damage Photo

MIDRA-7 Extract GPS Location

MIDRA-8 Dispatch Damage Report

MIDRA-9 Manage Damage Report

Damage Report Su...

Damage Report Su...

Location Identifica...

Report Dispatch

Engineer Report M...

To Do

To Do

To Do

To Do

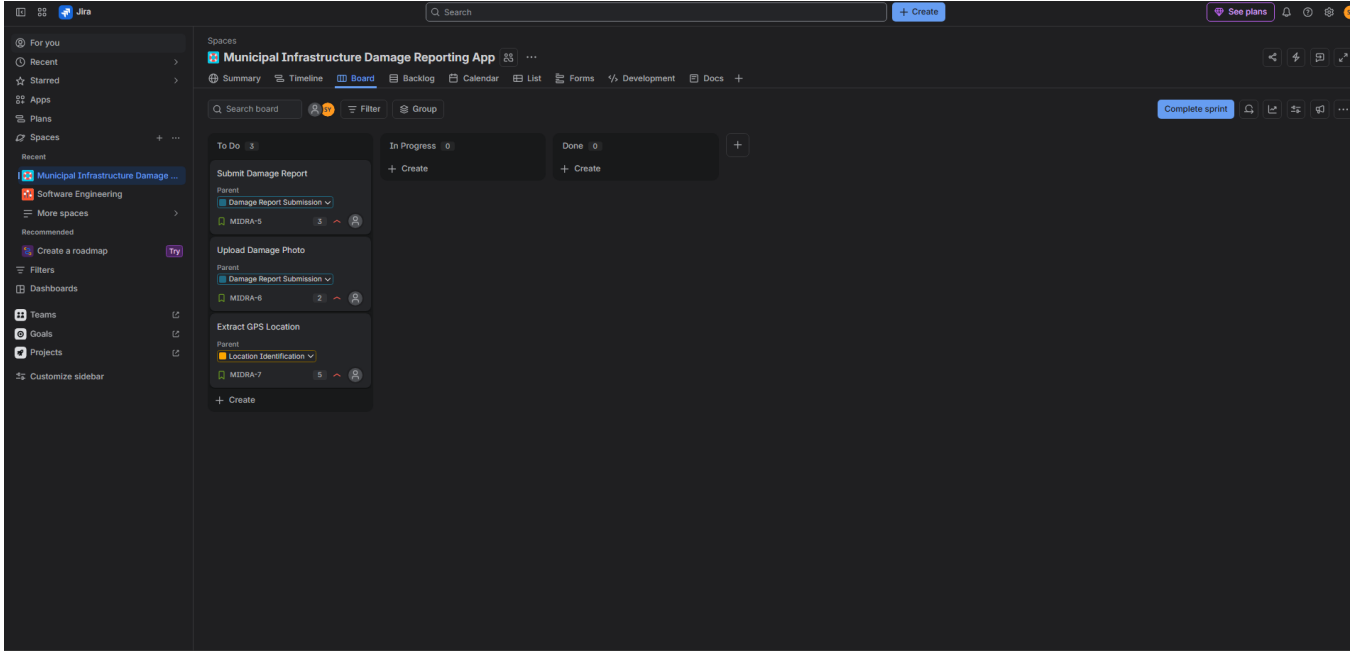
To Do

Create

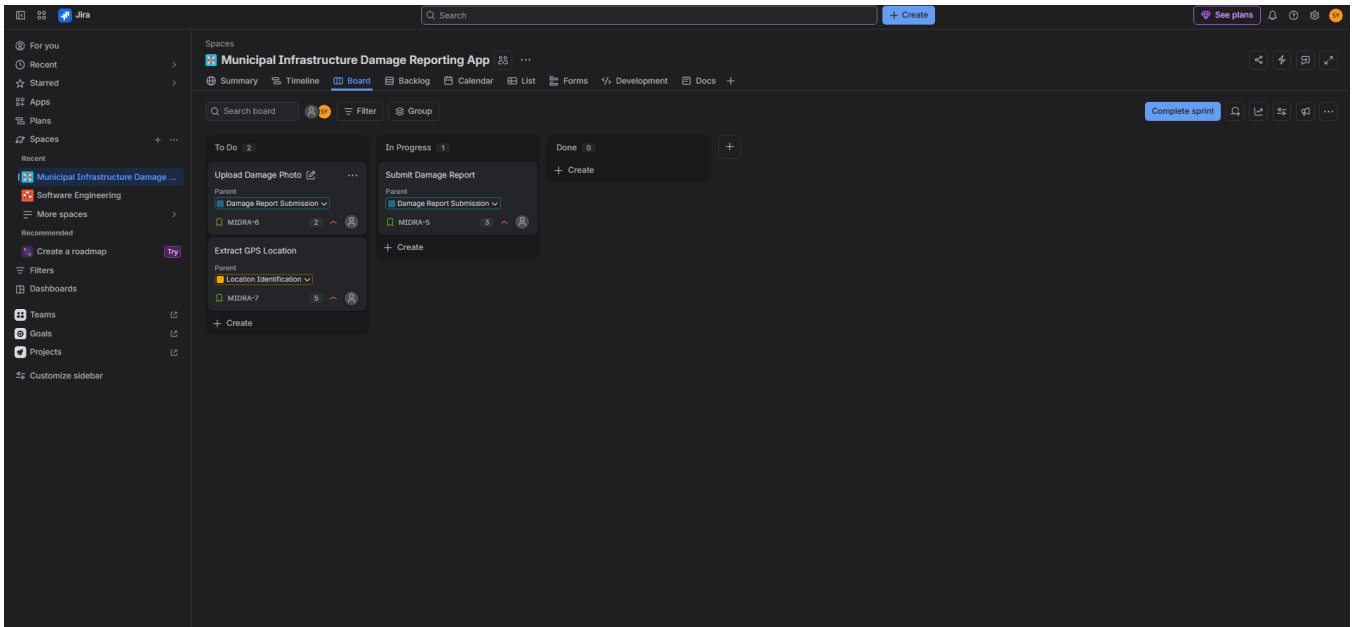
SPRINT 1

Sprint 1 included Submit Damage Report (3 points), Upload Damage Photo (2 points), and Extract GPS Location (5 points), giving a total planned scope of 10 story points. The board was used to simulate the movement of work from To Do through In Progress and finally to Done.

Initial Sprint 1 Board



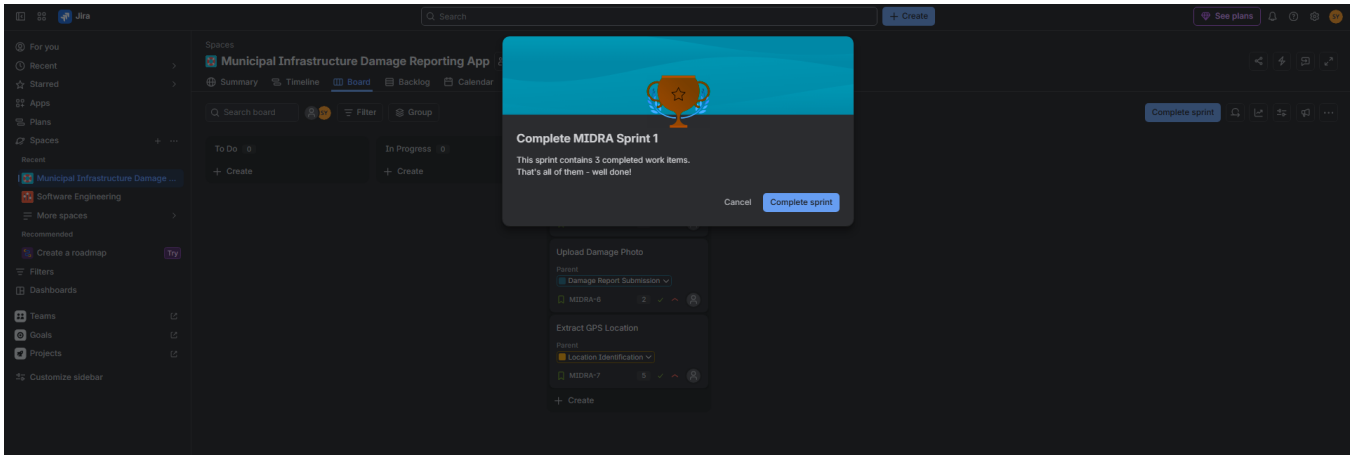
Sprint 1 Progress



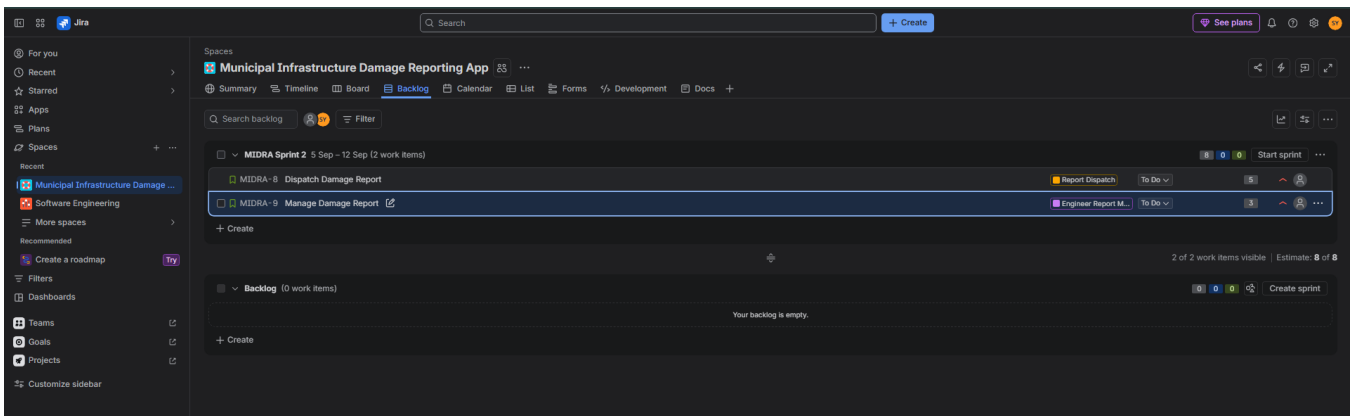
SPRINT 1 COMPLETION AND SPRINT 2 PLANNING

After the Sprint 1 stories were completed, the sprint was closed. The remaining two stories were then planned for Sprint 2: Dispatch Damage Report (5 points) and Manage Damage Report (3 points).

Sprint 1 Completion



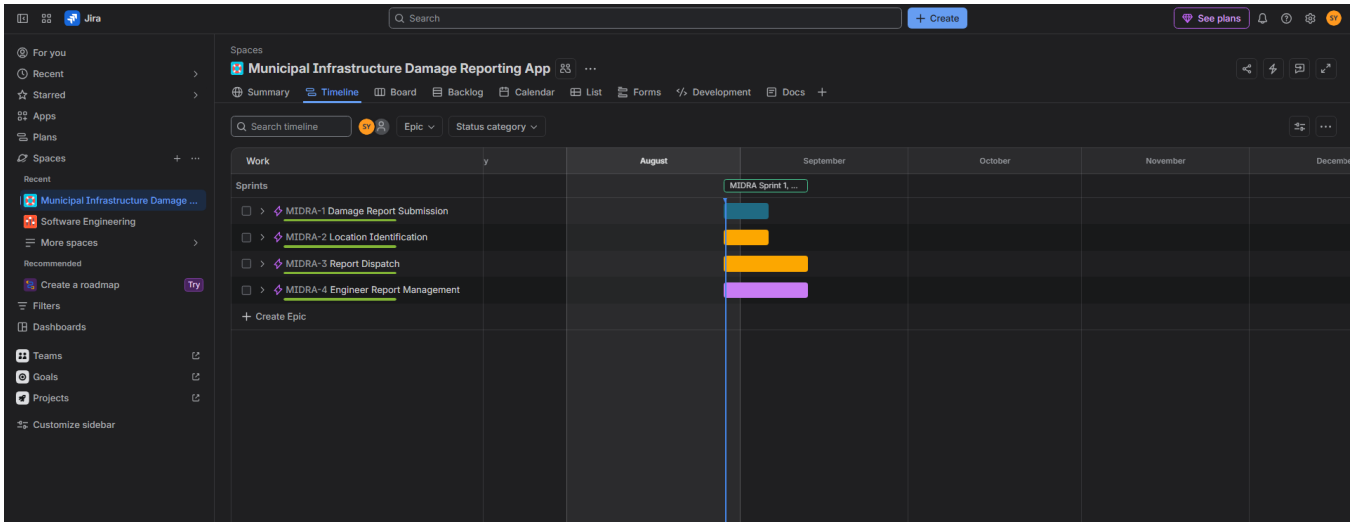
Sprint 2 Planning



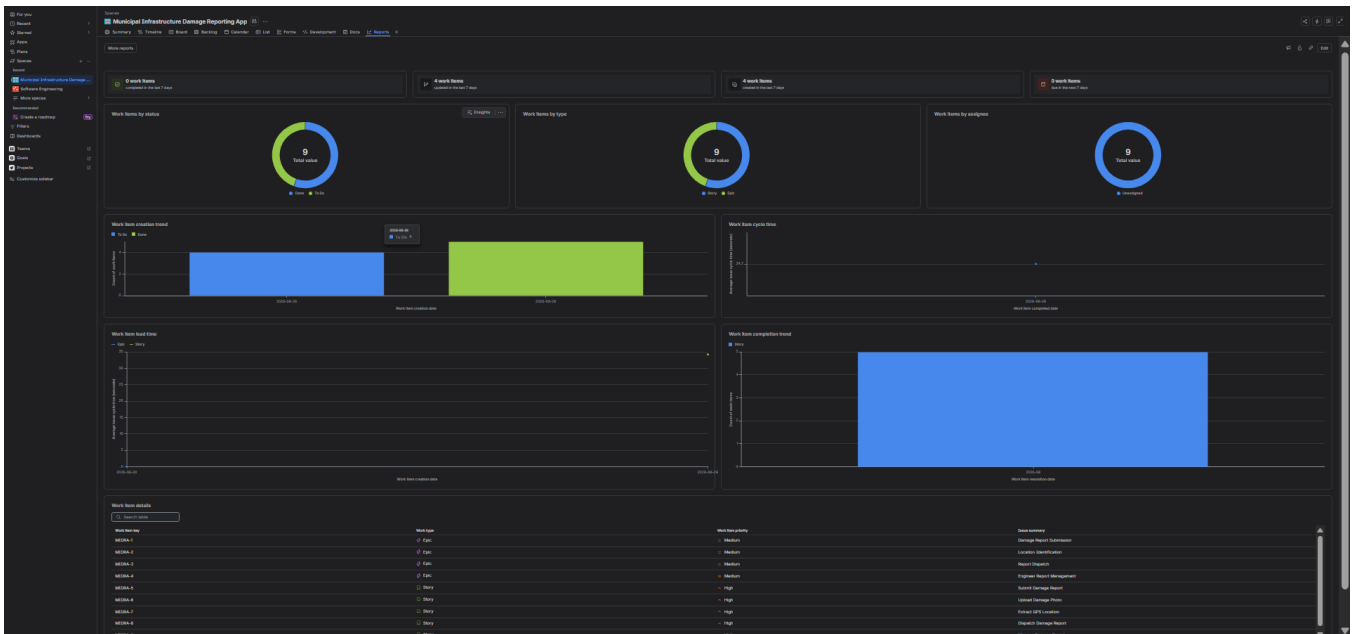
OVERALL PROJECT TRACKING

The Timeline provides a high-level view of the project structure and epic scheduling. The Jira Summary/Reports view provides a consolidated status snapshot, including work-item status, type, creation trend, completion trend, and work-item details.

Expanded Timeline



Jira Summary / Reports Dashboard



BURNDOWN CHART

The Sprint Burndown report was viewed for MIDRA Sprint 1 using Story points as the estimation field. The sprint covered 10 story points from the three planned stories. The burndown view represents remaining work over the sprint period and helps show whether work is being completed progressively.



REFLECTION QUESTIONS

1. Did your estimations reflect the actual effort?

Yes, the estimates were reasonably representative of the relative effort. Smaller activities such as photo upload were assigned fewer points, while GPS extraction was given a higher estimate because it involves location-related processing. The estimates were therefore useful for planning the sprint scope.

2. Was your backlog well-prioritized?

The backlog was kept focused on the main reporting flow. The stories were arranged according to the logical sequence of the application, starting with report creation and supporting data, followed by dispatch and engineer-side management.

3. How did your simulated sprint align with your plan?

The simulation followed the planned sprint allocation. Sprint 1 contained the first three stories and Sprint 2 contained the remaining two. The work items were moved through the board from To Do to In Progress and then to Done, which matched the intended Scrum workflow.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart made the amount of remaining work visible throughout the sprint. A step-down pattern indicates that completed stories reduce the remaining story points. It also gives a simple view of the amount of work planned and whether the team is progressing toward completing the sprint scope.

Conclusion

The Jira exercise demonstrated backlog creation, epic and story organization, story-point estimation, sprint planning, status transitions, sprint completion, timeline tracking, and burndown reporting for the Municipal Infrastructure Damage Reporting App.